

Finite Difference Methods for Solving the Black–Scholes Equation

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1 Introduction

The Black–Scholes partial differential equation (PDE) governing the price of a European option is

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (1)$$

where

- $V(S, t)$: option price
- S : stock price
- t : time
- r : risk free interest rate
- σ : volatility

The terminal condition for a European call option is

$$V(S, T) = \max(S - K, 0) \quad (2)$$

The PDE is solved backward in time from $t = T$ to $t = 0$.

2 Grid Discretization

Define

$$S_i = i\Delta S \quad (3)$$

$$t_n = n\Delta t \quad (4)$$

Denote

$$V_i^n = V(S_i, t_n) \quad (5)$$

3 Finite Difference Approximations

3.1 Time Derivative

Forward difference:

$$\frac{\partial V}{\partial t} \approx \frac{V_i^{n+1} - V_i^n}{\Delta t} \quad (6)$$

Backward difference:

$$\frac{\partial V}{\partial t} \approx \frac{V_i^n - V_i^{n-1}}{\Delta t} \quad (7)$$

Central difference:

$$\frac{\partial V}{\partial t} \approx \frac{V_i^{n+1} - V_i^{n-1}}{2\Delta t} \quad (8)$$

3.2 Space Derivatives

First derivative:

$$\frac{\partial V}{\partial S} \approx \frac{V_{i+1} - V_{i-1}}{2\Delta S} \quad (9)$$

Second derivative:

$$\frac{\partial^2 V}{\partial S^2} \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta S^2} \quad (10)$$

4 Why Central Difference is More Accurate

Taylor expand around S_i :

$$V_{i+1} = V_i + \Delta S V_S + \frac{\Delta S^2}{2} V_{SS} + \dots \quad (11)$$

$$V_{i-1} = V_i - \Delta S V_S + \frac{\Delta S^2}{2} V_{SS} + \dots \quad (12)$$

Subtracting:

$$V_{i+1} - V_{i-1} = 2\Delta S V_S + O(\Delta S^3) \quad (13)$$

Thus

$$V_S = \frac{V_{i+1} - V_{i-1}}{2\Delta S} + O(\Delta S^2) \quad (14)$$

Central difference has second order accuracy while forward and backward differences have only first order accuracy.

5 Explicit Scheme

Substituting finite differences into the PDE gives

$$V_i^n = a_i V_{i-1}^{n+1} + b_i V_i^{n+1} + c_i V_{i+1}^{n+1} \quad (15)$$

where

$$a_i = \frac{1}{2} \Delta t (\sigma^2 i^2 - ri) \quad (16)$$

$$b_i = 1 - \Delta t (\sigma^2 i^2 + r) \quad (17)$$

$$c_i = \frac{1}{2} \Delta t (\sigma^2 i^2 + ri) \quad (18)$$

6 Error Growth in Explicit Scheme

Suppose numerical errors are

$$\epsilon_i^n \tag{19}$$

Then

$$\epsilon_i^n = a\epsilon_{i-1}^{n+1} + b\epsilon_i^{n+1} + c\epsilon_{i+1}^{n+1} + \tau_i^n \tag{20}$$

where τ_i^n is truncation error.

If coefficients become negative, oscillatory errors may amplify.

For stability,

$$b_i \geq 0 \tag{21}$$

which implies

$$1 - \Delta t(\sigma^2 t^2 + r) \geq 0 \tag{22}$$

Using the largest value $i = M$,

$$\Delta t \lesssim \frac{1}{\sigma^2 M^2} \tag{23}$$

This is the stability restriction for the explicit scheme.

7 Implicit Scheme

The implicit formulation is

$$a_i V_{i-1}^n + b_i V_i^n + c_i V_{i+1}^n = V_i^{n+1} \tag{24}$$

Matrix form:

$$AV^n = V^{n+1} \tag{25}$$

Hence

$$V^n = A^{-1}V^{n+1} \tag{26}$$

8 Why Errors Get Controlled

Error propagation becomes

$$\epsilon^n = A^{-1}\epsilon^{n+1} \tag{27}$$

The matrix

$$A = \begin{bmatrix} b & c & 0 \\ a & b & c \\ 0 & a & b \end{bmatrix} \tag{28}$$

has diagonal dominance:

$$b > |a| + |c| \quad (29)$$

Therefore the inverse matrix remains bounded and does not significantly amplify errors.

9 Crank Nicholson Scheme

Crank Nicholson averages Explicit and Implicit methods:

$$\frac{V_i^{n+1} - V_i^n}{\Delta t} = \frac{1}{2} (LV^{n+1} + LV^n) \quad (30)$$

Define

$$\alpha = \frac{1}{4} \Delta t (\sigma^2 i^2 - ri) \quad (31)$$

$$\beta = -\frac{1}{2} \Delta t (\sigma^2 i^2 + r) \quad (32)$$

$$\gamma = \frac{1}{4} \Delta t (\sigma^2 i^2 + ri) \quad (33)$$

Final equation:

$$-\alpha V_{i-1}^n + (1 - \beta) V_i^n - \gamma V_{i+1}^n = \alpha V_{i-1}^{n+1} + (1 + \beta) V_i^{n+1} + \gamma V_{i+1}^{n+1} \quad (34)$$

Matrix form:

$$AV^n = BV^{n+1} \quad (35)$$

10 Comparison

Method	Stability	Accuracy	Matrix solve
Explicit	Conditional	First order	No
Implicit	Unconditional	First order	Yes
Crank Nicholson	Unconditional	Second order	Yes

11 Conclusion

Explicit methods are simple but can become unstable when errors grow over time. Implicit methods control these errors through the matrix inverse A^{-1} , while Crank Nicholson combines the advantages of Explicit and Implicit schemes and provides higher accuracy.